Level 2: Wi-Fi Setup and Secure Telemetry

Scenario: The school installs the device itself and needs a local status page plus reliable cloud storage of every reading.

Purpose: Make the device configurable after deployment and transmit trustworthy, unique telemetry to the backend.

Main requirements

- When no network is saved, start the setup access point and a setup page; list nearby Wi-Fi networks and signal strengths.
- Provide setup fields for Wi-Fi password, Room ID and device token; store configuration in flash/NVS.
- Serve a local status page from the ESP32-S3 with room, local time, sensor values, status, flame, door, last card UID, signal strength, chip temperature, firmware and connection state.
- Store local page files in LittleFS and refresh the page automatically at least every 2 seconds.
- Reconnect automatically after Wi-Fi loss without restarting; after power restoration reconnect without repeating setup.
- Return to setup mode within 15 seconds after invalid credentials or an unreachable network; allow BOOT held for 5 seconds or reset command to clear saved settings.
- Keep Level 1 tasks active during all network operations.
- Send telemetry securely; use NTP timestamps and a monotonically increasing sequence number that is never reused after restart.
- Reject duplicate/replayed sequence numbers at the server; show device offline within 30 seconds of silence and online after reconnection.
- Keep passwords, keys and tokens out of code and the repository; maintain a README with build and run instructions.

Where the result must appear

- Serial Monitor: setup password and network events; local page: device fields and connection state; database: telemetry, CARD_TAP and BOOT events.

Live judging focus

- Complete setup on the judge hotspot; show local addresses; show telemetry records arriving at the specified interval; reject replay; show offline and automatic recovery.

Video evidence

- Disconnect hotspot for 20 seconds; hold BOOT for 5 seconds and test wrong hotspot credentials; stop backend for 20 seconds and show records resume without device reset.

Write-up evidence

- Setup and reconnection design; mDNS name; architecture diagram covering device, server, database, AI/weather services and dashboard; replay prevention; TLS certificate validation; secure secret handling and README.

